Scientists Find Dinosaur Blood That Survived 115 Million Years

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Scientists have found traces of dinosaur blood inside bones that are 115 million years old. The discovery shows that biological molecules can survive far longer than researchers once believed. An international team led by Curtin University in Australia studied the limb bones of Aratasaurus museunacionali, a meat-eating dinosaur found in Brazil. Inside tiny passages that once held blood vessels, the team found **fossilized** red blood cells. Chemical tests revealed heme molecules, parts of **hemoglobin**, the protein that lets red blood cells carry oxygen. The team also found amino acids linked to **collagen**, a structural protein found in bones and other connective tissues. This is the first time scientists have found chemical proof of **hemoglobin**-derived molecules inside **fossilized** dinosaur red blood cells. So how did the blood survive? Part of the answer is a mineral called **hausmannite**, which helped stabilize the cells and protect their molecules. The dinosaur was probably buried quickly in **sediment** with little oxygen, conditions that may have slowed the usual breakdown of biological material. Professor Kliti Grice, who led the study, called the find a landmark discovery.

Word Treasure

fossilized -- Turned into stone over a very long time.
hemoglobin -- The protein in red blood cells that carries oxygen around the body.
collagen -- A strong protein that helps hold bones and other body parts together.
hausmannite -- A mineral that helped protect the old blood cells.
sediment -- Sand, mud, or small bits of rock that settle at the bottom of water or land.

Background you need

Dinosaurs lived on Earth long ago, and their bones can turn into fossils over a very long time.

Blood is made of tiny parts, like red blood cells that carry oxygen around the body, and those parts usually break down quickly after an animal dies.

Scientists study old bones to learn how living things worked, and this find shows that tiny pieces of blood can sometimes survive inside bones for a very long time.

Break it down

WHO: An international team led by Curtin University in Australia, including Kliti Grice

WHAT: The team found traces of dinosaur blood, including fossilized red blood cells and heme molecules, inside old limb bones

WHERE: Inside the limb bones of Aratasaurus museunacionali, a meat-eating dinosaur found in Brazil

WHY: The blood may have survived because hausmannite helped protect it and the dinosaur was buried quickly in sediment with little oxygen

Why it matters

It shows that tiny clues from long ago can still teach us how living things worked, just like clues in your own body help us learn how you stay healthy.

Test what you remember

1. What did scientists find inside the dinosaur bones?

- (A) A pile of old leaves
- (B) A piece of a meteor
- (C) A baby dinosaur
- (D) Traces of dinosaur blood

2. What is hemoglobin?

- (A) The protein that lets red blood cells carry oxygen
- (B) A type of rock
- (C) A kind of dinosaur food
- (D) A tool for digging up fossils

3. Where were the fossilized red blood cells found?

- (A) On the top of a tall mountain
- (B) Inside tiny passages that once held blood vessels
- (C) In a museum gift shop
- (D) In the sea

4. What mineral helped protect the blood cells?

- (A) Quartz
- (B) Hausmannite
- (C) Sand
- (D) Salt

5. Which dinosaur's limb bones did the team study?

- (A) A flying pterosaur
- (B) A woolly mammoth
- (C) Aratasaurus museunacionali
- (D) A giant sea turtle

6. How was the dinosaur probably buried?

- (A) In ice on a cold mountain
- (B) Quickly in sediment with little oxygen
- (C) Under a pile of leaves in a forest
- (D) Slowly in dry sand with lots of air

Answer key (try the quiz first!): 1: D 2: A 3: B 4: B 5: C 6: B

Questions to think about

1. Kliti Grice: Called the find a landmark discovery.

MY THOUGHTS (at least 20 words):